

Stephanie Villela

San Diego, CA • 956-312-6481 • stephanie.villelap1@gmail.com • linkedin.com/in/sv-35606427b

PROFESSIONAL SUMMARY

AI Engineer at ASML architecting agentic AI systems, GPU-native scientific computing pipelines, and multi-agent platform evaluations across NVIDIA ALCHEMI, Databricks, Azure, Google Vertex AI, and Mistral. B.S. Computer Science, Boston College (2026), with a track record of turning cutting-edge AI research into production tools that save engineering and technician time.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, JavaScript, Apache Spark, Dash / Plotly, React, Flask, Figma

AI / ML: PyTorch, TensorFlow, XGBoost, OpenCV, OpenAI API, MLIP Models (MACE, TensorNet), Time Series Forecasting (Prophet), Anomaly Detection, Agentic Systems

Platforms: NVIDIA ALCHEMI, Databricks Mosaic AI, Azure AI Foundry, Google Vertex AI, NVIDIA NIMs, Mistral AI Studio, ServiceNow, Microsoft Copilot Studio

EDUCATION

Boston College — Boston, MA

B.S. Computer Science, May 2026

WORK EXPERIENCE

AI Engineer

May 2026 – Present

ASML – AI Enablement Team

- Built an agentic materials-screening pipeline on NVIDIA ALCHEMI, a GPU-native computational chemistry environment, to test oxide materials for tin adsorption — autonomously benchmarking 2,000+ materials across multiple MLIP models (MACE, TensorNet PBE, TensorNet r2SCAN) and reporting results, delivering significant time and cost savings over manual testing.
- Automated SEM defect detection using OpenCV to replace manual technician classification of 5 defect types across 1,000+ real orifice images: engineered an ellipse-fitting boundary model to compute per-pixel radial deviation, then trained XGBoost models to predict defect probability, delivered via a Streamlit dashboard pairing each prediction with an AI-generated visual and mathematical explanation and reducing manual review time.
- Onboarded and supported multiple users on a GPU-accessible scientific computing platform, acting as de facto data scientist to ensure data pipelines, storage, and scripts ran reliably across varied research use cases.

AI Engineer Intern

May 2025 – May 2026

ASML – AI Enablement Team

- Built Prophet-based time series forecasting models across Databricks projects, analyzing a year of historical cluster usage data to power an AI-driven dashboard — prototyped in Dash / Plotly and productionized as a Databricks App — that monitors cost and automatically flags over-budget usage.
- Designed an agentic KPI retrieval application, delivered as a Flask app, that for any machine KPI query returns the past month's functional summary, defining code, dependent KPIs, data lineage, and notable recent shifts with visualizations — replacing reliance on institutional knowledge to validate machine health.
- Built an AI use-case recommendation platform that guides users to the optimal enterprise AI tool for their agentic initiatives — Databricks Mosaic AI, Google Vertex AI, Mistral AI Studio, NVIDIA NIMs, Azure AI Foundry, and Microsoft Copilot Studio — based on internal data context and security requirements.

AI/ML Engineer Intern

May – Aug 2024

Mindhome Inc

- Integrated Meta's I-JPEG model using PyTorch and OpenCV for AI-powered home security surveillance, targeting real-time detection of difficult-to-identify objects.
- Benchmarked model performance against TensorFlow baselines across lighting conditions and object categories to assess production readiness for residential deployment.

PROJECTS

Agentic Trading Bot — Claude-Powered Autonomous Trading Agent

2025 – Present

- Built an autonomous trade execution agent using the Claude API connected via an MCP integration to Robinhood, enabling the agent to monitor market signals and independently execute trades; prototyped and iterated on the workflow in Claude Cowork.